

Inversion in Russian, Smuggling, and Leapfrogging

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Abstract: Russian is a canonically-SVO language with relatively free word order (Bailyn 1995). As others have shown, OVS word orders for transitive clauses involve A-movement of the preverbal object (Bailyn 2004; Pereltsvaig 2021, a.o), while the fronted object of OSV sentences lacks A-properties and is derived via $\bar{A}$ -movement. In this squib, we account for these facts under a Minimalist analysis of inversion as smuggling (Collins 2024; Storment 2025b), showing that smuggling accounts for the OVS word order and the A-properties of the fronted object. We contrast the smuggling analysis with an alternative, leapfrogging, which we show fails to account for the OVS-OSV asymmetries regarding A-movement, as well as introducing a general theoretical issue of unrestrictiveness.

Keywords: inversion, OVS, Russian, smuggling, leapfrogging

1. Introduction

Russian, a canonical SVO language with a relatively free word order, allows an OVS word order in transitive clauses.

- (1) a. Ivan čita -et knig -u. (SVO) (Bailyn 2004:41)
Ivan.NOM read-3SG.PRES book-ACC
b. Knig-u čita -et Ivan. (OVS)
book-ACC read-3SG.PRES Ivan.NOM
'Ivan is reading a book.'

Generally, this construction is analyzed as involving A-movement of the accusative object DP to the position of structural subject (i.e., that it is in Spec,TP) (King 1995; Bailyn 2001; 2003; 2004; 2012; Titov 2012; 2013; Pereltsvaig 2021, a.o), though this point is not without disagreement (Erechko 2003; Slioussar 2011; Wiland 2013; Ionin & Luchkina 2018). We present several empirical diagnostics from the literature confirming the location of the accusative DP in an A-position in (1b).

The word order OSV is also possible, as shown in (2). But with that order, it is possible to show that the preposed object is no longer in Spec TP, but rather higher in the left periphery.

- (2) Èt -u knig -u Ivan čita -et. (OSV) (Bailyn 2004:6)
this-ACC book-ACC Ivan.NOM read-3SG.PRES
'This book, Ivan reads.'

This difference in the A-status of the preposed accusative object DP in (1b) versus (2) reveals several interesting insights into the nature of inversion. First, the data lend themselves exceptionally well to an analysis of inversion as smuggling (Collins 2005; 2024; Storment 2025a; b; c). Second, these data suggest that any analysis of A-movement of an internal argument to a position over an external argument that involves leapfrogging (Bobaljik 1995; McGinnis 1998; Legate 2014; Keine & Zeijlstra 2024, a.o) is not on the right track, as leapfrogging cannot account for the correlation between A-movement and inversion (the verb preceding the external argument).

2. A-movement in OVS (and lack thereof in OSV)

It is well-established that Russian OVS clauses involve A-movement of the accusative internal argument DP (Bailyn 2004; Pereltsvaig 2021, a.o). For example, the OVS word order obviates weak crossover, while the OSV order does not. The obviation of weak crossover is a clear property of A-movement. See the following examples from Pereltsvaig (2021:7-8).

- (3) *Eë_i xozjajka otremonirova-la každuj-u kvartir -u_i. (SVO)
 its owner.NOM renovate -3FSG.PAST every -ACC apartment-ACC
int: ‘Every apartment was renovated by its owner.’
- (4) Každuj-u kvartir -u_i otremonirova-la eë_i xozjajka. (OVS)
 every -ACC apartment-ACC renovate -3FSG.PAST its owner.NOM
 ‘Every apartment was renovated by its owner.’ (Titov 2012:16, 19; 2013:36)
- (5) *Každuj-u kvartir -u_i eë_i xozjajka otremonirova-la. (OSV)
 every -ACC apartment-ACC its owner.NOM renovate -3FSG.PAST
int: ‘Every apartment was renovated by its owner.’

In (3), the quantifier phrase does not c-command the bound variable at Spell-Out, giving rise to a classic WCO effect. In (4), the object has been A-moved to Spec TP, with no WCO violation. In (5), the object has been A'-moved to the left periphery giving rise to a WCO violation.

We can compare the grammaticality of OVS in (4) to another derivation involving A-movement of the internal argument: passives. In (6), just like in (4), movement of the quantifier phrase over the external argument does not give rise to a WCO violation.

- (6) Každaja kvartira_i byl -a otremonirova-na eë xozjajk-oj_i. (Passive)
 every.NOM apartment.NOM was-3FSG renovate -PASS its owner -INSTR
 ‘Every apartment was renovated by its owner.’

Additionally, we see that reciprocals and reflexives contained within the external argument can be bound by the fronted object in OVS clauses, but not OSV clauses. See the following examples from Pereltsvaig (2021:4-5).

- (7) *Vystrely drug_i druga ubi-li duèljant-ov_i. (SVO)
 shots.NOM each other kill-3PL.PST duelists -ACC
int: ‘The duelists were killed by each others shots.’
- (8) Duèljant-ov_i ubi -li vystrely drug_i druga. (OVS)
 duelists -ACC kill-3PL.PAST shots.NOM each other
 ‘The duelists were killed by each other’s shots.’
- (9) *Duèljant-ov_i vystrely drug_i druga ubi -li. (OSV)
 duelists -ACC shots.NOM each other kill-3SG.PAST
int: ‘The duelists were killed by each other’s shots.’

In the cases of WCO and reciprocal binding, OVS stands out from SVO and OSV as it is the only case that involves A-movement of the internal argument over the subject, thus patterning

with passives. There are further diagnostics for A-movement in OVS (and the lack thereof in OSV) that we do not show here for the sake of brevity. For example, Bailyn (2004:29) shows that the distribution of idiom chunks provides further support for these claims.

These observations give rise to the following generalization (due to Pereltsvaig 2021):

(10) *Pereltsvaig's Generalization* (for Russian)

OVS: O undergoes A-movement.

OSV: O undergoes $\bar{A}$ -movement.

This generalization holds not only for Russian, but is also applicable to OVS and OSV derivations in English (Culicover & Levine 2001; Murphy 2022; Storment 2025b) and Bantu (Demuth & Mmusi 1997; Diercks 2017; Storment 2025c) as well. Thus, there is a general association with A-movement and inversion (the verb preceding the external argument). We propose the following much stronger cross-linguistic generalization:

(11) *Inversion Generalization* (Universal)

If an SVO language has an alternation between OVS and OSV word orders, then:

OVS: O undergoes A-movement.

OSV: O undergoes $\bar{A}$ -movement.

We know of no exceptions to this generalization, which is elegantly explained by the analysis of inversion as smuggling presented in the following section.

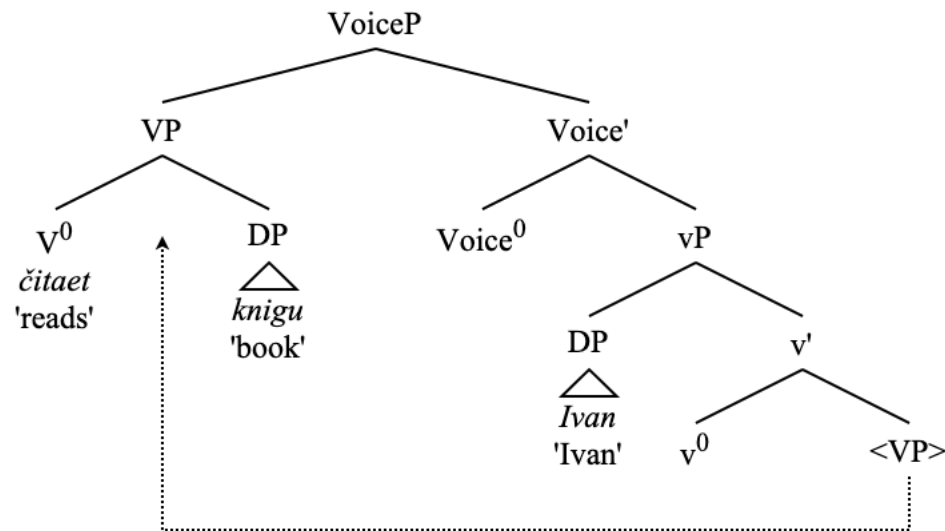
3 Inversion as smuggling

In light of the data above, an analysis of the OVS construction must account for the fact that an internal argument has undergone A-movement to a position above the external argument. In addition to this apparent minimality violation, the fact that the verb precedes the external argument must also be explained. Both of these facts can be handled – and tied together – with smuggling (Collins 2005; 2024; Storment 2025b).

Smuggling involves the movement of a given element within a larger constituent, thus making that element a viable goal for processes that would normally be blocked by an intervener, as the larger constituent moves to a position above the intervener. Following Collins (2005; 2024) and Storment (2025a; b; c), we assume that smuggling is mediated by a functional projection VoiceP, whose role is related to the licensing of arguments via the mapping of those arguments onto A-positions. This treatment of VoiceP is a notable departure from the view of VoiceP as an applicative head that introduces an external argument (Kratzer 1996; Pylkkänen 2008, a.m.o).

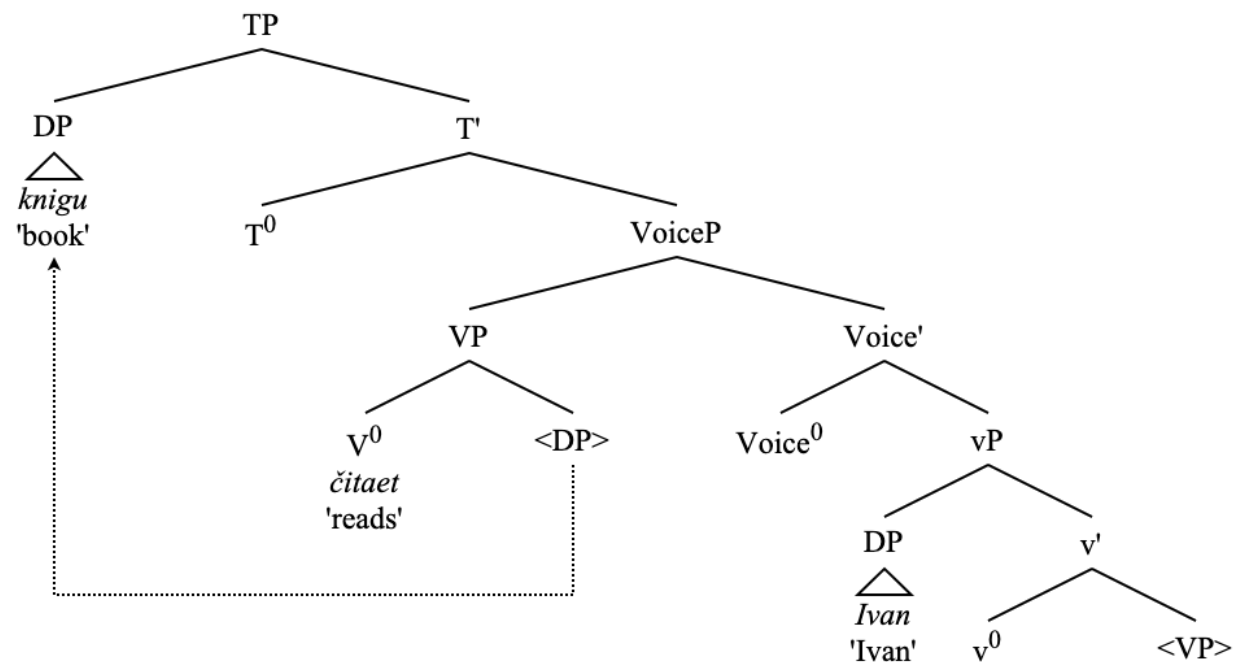
In these Russian OVS clauses, the smuggling derivation begins with the VP containing the internal argument moving to Spec,VoiceP. For a discussion on the featural motivation of this movement, see Storment (2025b). For an explanation of this VP movement connected to the labeling algorithm, see Blümel & Collins (2025). See the following structure for example (1b).

- (12) VP movement to Spec, VoiceP



At this point, a phi-probe on T^0 is merged, and the internal argument raises to Spec TP, satisfying the EPP feature of T^0 :

- (13) The internal argument moves to Spec, TP



Lastly, T^0 agrees with the post-verbal subject, triggering nominative Case checking (see Collins 1997 on agreement with post-verbal subjects, and Storment 2025 for much additional discussion).

It is clear, then, that Spec, TP in Russian is not unilaterally associated with nominative Case, like it is in English. This property of Russian Spec, TP is further confirmed by the fact that dative,

instrumental, and accusative DPs outside of OVS clauses can also A-move to Spec,TP (Bailyn 2004; 2012).

In (13) the internal argument has moved over the external argument by smuggling. In effect, smuggling allows A-movement of one DP over another without violating any of the classical locality constraints, including Relativized Minimality (Rizzi 1990), the Minimal Link Condition (Chomsky 1995) and the Phase Impenetrability Condition (Chomsky 2000).

The VP movement in the smuggling derivation also naturally explains the fact that the verb appears in a position before the external argument (hence, the VS in OVS), which is necessary as Russian notably lacks v-to-T verb raising (Bailyn 1995a; b; Babyonyshev 1996; Brown 1999; Pereltsvaig 2021, a.o; cf. Bailyn 2004).

Pereltsvaig (2021) rejects a VP movement analysis of post-verbal subjects in inversion constructions based on the so-called Fronted VP Scope Freezing effect. For the sake of brevity, we do not discuss her analysis here. But we note that the movement of VPs that give rise to scope freezing involves $\bar{A}$ -movement to the left periphery. The VP movement we postulate for smuggling, is neither A-movement nor $\bar{A}$ -movement, but is purely phase internal movement.

4. Against Leapfrogging

An alternative account of the facts presented above involves the object undergoing A-movement to an outer specifier of vP (reminiscent of the leapfrogging approach of Legate (2014:55), as illustrated below:

(14) [_{vP} Obj [_{v'} Subj [_{v'} v VP]]

On this approach, the object DP raises to the outer-specifier of vP, in order to move past the subject in the inner specifier of vP. By hypothesis, this movement does not violate Relativized Minimality, because the inner and outer specs are equidistant. Lastly, the object moves to Spec,TP (as documented by Bailyn 2004 and Pereltsvaig 2021).

The problem with this approach is that it has no way to explain the correlation between A-properties of the fronted object and inversion (the verb preceding the subject). In other words, only when there is inversion of the main verb and the subject, does the object come to have A-properties (for binding and WCO). If the derivation in (14) were possible (with A-movement of the object), then the prediction would be that preposing of the object in OSV word orders (see (5) and (9)) should show A-movement properties. We conclude that leapfrogging derivations must be excluded, since they lead to overgeneration.

In general, there are two large problems with leapfrogging: (a) It is unrestrictive. If one chooses to adopt it, there are no more locality constraints on A-movement; they can all be circumvented. (b) It requires a severe complication of locality theory, since it presupposes that multiple specifiers are equidistant (even though the outer one clearly asymmetrically c-commands the inner one). The smuggling account handles the data presented in this paper without assuming such theoretical complications.

5. Conclusion

In this paper, following Bailyn (2004), Pereltsvaig (2021), and others, we have shown that Russian OVS inversion clauses are derived by A-movement of the internal argument to Spec,TP, and that fronted objects in OSV sentences do not share the A-properties, and are thus derived via $\bar{A}$ -movement.

We have shown that these facts, along with the position of the verb in OVS clauses, naturally follow from an account of inversion as smuggling. In smuggling, A-movement of the object to Spec,TP corresponds with the verb preceding the external argument because the VP moves to Spec,VoiceP. Without this movement, there is no possibility of A-movement of the object, which explains how there are no A-properties of the object in OSV clauses. In addition to making the correct predictions, the smuggling analysis proves superior to a leapfrogging analysis because it is not as unrestrictive, and because it does not assume theoretical complications such as equidistance and multiple specifiers.

As mentioned in (11), this contrast between OVS and OSV clauses regarding A-movement holds in other languages that allow inversion, such as English and various Bantu languages. Thus, the analysis of inversion as smuggling naturally extends to cover data from several other languages as well.

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